

Curriculum Vitae of Jie Yuan (FRAeS, CEng, FHEA)

Employment History

- 07/2023- present: Assistant Professor in Aerospace Design and Optimization, Department of Aeronautics and Astronautics, University of Southampton, UK.
- 12/2020-06/2023: Lecturer in Aerospace Engineering, Department of Mechanical and Aerospace Engineering, University of Strathclyde, Glasgow, UK.
- 04/2017-11/2020: Research Associate in Vibration University Technology Centre, Department of Mechanical Engineering, Imperial College London, UK. Project: EPSRC SYSDYMAT (2019-2022) and ATI GEMinIDS (2017-2019)
- 10/2015-03/2017: Research Engineer, Airbus UK & Cranfield University, Bristol, UK.

Education History

- 07/2012-02/2016: PhD in Aerospace Engineering at the University of Bristol under the supervision of Prof. Fabrizio Scarpa.
- 10/2010-02/2012: MSc in Integrated Aerospace System Design (Top 3), University of Bristol, UK.
- 09/2006-07/2010: BE in Mechanical Engineering (First Class, 5/200), Ningbo University, Zhejiang, China.

Membership & Certification & Awards

- [Fellow of Royal Aeronautic Society](#)
- [Research Fellowship in Royal Academy of Engineering/ Leverhulme Trust](#)
- [Committee member in Propulsion Specialist Groups in RAeS](#)
- Scientific Committee Member, International Structural Joint Community
- [Fellowship of Advance Higher Education](#)
- [Chartered Engineer, UK Engineering Council](#)
- Youth Member of Editorial Board, [Theoretical and Applied Mechanics Letters](#) (Elsevier)
- Member of the American Society of Mechanical Engineers (ASME)
- Member of the Society for Experimental Mechanics (SEM)
- [Best paper Award in Technical Division in Nonlinear Structures and Systems, IMAC, 2021](#)

Fellowship & Research Grants (PI)

- Co-PI (only one), Innovative UK, DDT: Digital Design Transformation, Innovative UK, 2025-2028, 1.62M
- PI, EPSRC Postdoctoral Fellowship (Dr. Michael McGurk), Invariant manifold based reduced order modelling for nonlinear aeroelastic system, 2025-2027
- PI, Royal Academy of Engineering Fellowship, Multi-scale Friction Interface Design for Resilient Dynamical Systems, 2023-2024, 58 K
- PI, Royal Society Research Grant, Controlling limit cycle oscillations for complex aerospace systems, 2024-2025, 20 K
- PI, Royal Society International Exchanges, Multi-functional nonlinear tuned vibration energy harvester for intelligent rotating systems, 2022-2025, 12 K
- PI, Impact Acceleration Award, EPSRC, Uncertainty Quantification of Novel Aeroelastic Systems using Advanced Composite Materials, 2022, 15 K
- PI, Small Research Award, Royal Society of Edinburgh, Multi-scale thermomechanical analysis of friction involved complex dynamic systems, 2021-2022, 5K
- PI, Saltire Workshop Award, Royal Society of Edinburgh, Uncertainty quantification for nonlinear dynamic systems with friction interface, 2022-2023, 10K

PhD studentship (PI or Awardee)

- PI, EPSRC iCASE with Rolls Royce plc, AI-enable stochastic load identification due to fan blades off, University of Southampton 2025-2029 (160K)

- PI, Research Excellent Award, EPSRC DTP, Robust nonlinear aeroelasticity tailoring using composite materials, University of Strathclyde, 2021-2025 (85K)
- PI, Strathclyde International Partnership, Mutual effects between friction and vibration across scale and physics, University of Strathclyde, 2022-2025 (60K)
- Awardee, Rolls Royce Studentship, Reduced order modelling of mistuned bladed disc systems, University of Bristol, 2012-2016 (60K)

Invited Talks

- Yuan, J. “Friction damping for turbomachinery: a comprehensive review of modelling, design strategies, and testing capabilities”, March 2025, University of Cambridge, UK
- Yuan, J. “Data driven Bayesian identification for complex aeroelastic systems”, Oct 2024, Imperial College London, UK
- Yuan, J. “Influence of mesoscale friction interface geometry on nonlinear dynamics”, 26 Oct 2022, National Institute for Research in Digital Science and Technology (INRIA), Rennes, France
- Yuan, J. “High fidelity modelling of jointed structures with friction interface in aero engine applications”, Digital Twins Workshop on Numerical and Experimental Studies of Structural Joints, June 2021, University of Swansea.
- Yuan, J. “An Overview of Reduced Order Modelling Techniques for Dynamic Analysis of Jointed Structures with Localized Nonlinearities due to the Contact Friction”, 23rd Blade Mechanics Seminar, Sep 2018, Winterthur, Switzerland

Academic Engagement

- EPSRC Peer Review College, 2024-Now
- Session Chair in Computational Engineering Design, Annual Rolls Royce Doctoral Symposium, June/July 2024-2026
- Scientific committee member, conference on “[Recent Advances in Non-Smooth Dynamics](#)”, Exeter, UK, December 2025
- Workshop organizer, “[Multiscale Structural Dynamics of Uncertain Frictional Interfaces](#)”, 8 Jun 2023
- Leading Guest Editor, ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems: Part B. (ISSN SI058B), Special Issue on “[Uncertainty Quantification & Management in Nonlinear Dynamical Systems in Aerospace Engineering](#)”, 2022-2023
- Leading Guest Editor, Applied Mechanics (ISSN 2673-3161), Special Issue on “[Vibrations and Friction: Theoretical, Experimental and Numerical studies](#)”, 2021-2024
- Session Chair, Third and Fourth International Nonlinear Dynamics Conference, June 2023 & 2025
- Peer Reviewer (8-10 papers a year), Mechanical System and Signal Processing, AIAA Journal, Journal of Sound and Vibration, Nonlinear Dynamics, Royal Society Proceeding A
- Departmental seminar and conference organiser, Department of Aeronautics and Astronautics, University of Southampton

Teaching Activities

- Digital Aerospace, UG, Lecturer and Co-module lead, University of Southampton, 2024-Now, 250 students
- Static and Dynamics, PG, Lab lead, University of Southampton, 2024-Now, 250 students
- Aeroelasticity, PG, Lecturer, University of Southampton, 2025-Now, 20 students
- Aerospace Propulsion, UG, Lecturer and module lead, University of Strathclyde, 2020-2023, 120 students
- Aero Design 1, UG, Lecturer, University of Strathclyde, 2020-2023, 150 students
- Program Director, MSc in Advance Mechanical Engineering, University of Strathclyde, 2020-2023, 40 students
- Postgraduate Internal and External Examiner, University of Strathclyde and University of Bristol

PhD Supervision experience

- Mr. Mingzhi Huang, PhD, Main Supervisor, University of Southampton, 2025-Now
Project: Design and optimisation of nature inspired architecture structures for passive dynamic control
- Ms. Giovanna Vilela, PhD, Main Supervisor, University of Southampton, 2025-Now
Project: AI-assisted Robust Aero-engine Structural Analysis Due to Fan Blade-off
- Mr. Peiyu Wang, PhD, Main Supervisor, University of Southampton, 2024-Now

Project: Robust data driven modelling for complex nonlinear and high dimensional dynamic systems

- Mr. Thomas Tsing, PhD, Co-supervisor, University of Southampton, 2024-Now

Project: Convolutional Generative Adversarial Network for Rapid Engineering Design

- Mr. Ahmad Algara, PhD, Main Supervisor, University of Strathclyde, 2022-Now

Project: Multi-scale modelling for friction induced vibration systems

- Mr. Harry Leitch, PhD, Main Supervisor, University of Strathclyde, 2022-Now

Project: Robust optimisation of tow steering composite for aerospace applications

- Dr. Michael McGurk, PhD, Main Supervisor, University of Strathclyde, 2021-2025

Project: [A Robust Data-Driven Bayesian approach for Complex Nonlinear Aeroelastic System Identification](#)

- Dr. Yekai Sun, PhD, Co-supervisor, Imperial College London, 2017-2021

Project: [Nonlinear modal analysis for blisks with friction dampers](#)

Selected Publication list

[Google Scholar](#), citation 1163, H-index 21 and i10-index 31

1. Algara, A., Cabboi, A., & Yuan, J. (2026). Passive control through surface roughness of local stability boundaries for friction-induced vibration. *International Journal of Non-Linear Mechanics*, 105419. <https://doi.org/10.1016/j.ijnonlinmec.2026.105419>
2. Wang, P., Toal, D., & Yuan, J. (2026). CNN-assisted robust design of an intentionally mistuned frictional damper for bladed-disk systems. *ASME-Journal of Turbomachinery (Accepted)*
3. Yang, D., Lu, Z., Yuan, J. et al. An event-driven time-domain sensitivity method for optimizing parameters of vibro-impact NES cells to suppress multi-mode vibration. *Journal of Sound and Vibration* <https://doi.org/10.1016/j.jsv.2026.119664>
4. Shao, K., Wang, T., Yuan, J., Dong, X., & Yang, J. (2026). Vibration Power Flow and Dynamic Interactions in Nonlinear Mistuned Bladed Disk System. *International Journal of Mechanical Sciences*, 111190. <https://doi.org/10.1016/j.ijmecsci.2026.111190>
5. Huang, M., Yuan, J., Leitch, H., & Scarpa, F. (2025). Analysis and optimisation of cellular Kirigami wingbox (CKW) structures for enhanced aeroelastic performance. *The Aeronautical Journal*, 1-17. [10.1017/aer.2025.10073](https://doi.org/10.1017/aer.2025.10073)
6. Leitch, H. J., Stodieck, O., & Yuan, J. (2025). The influence of coupled thickness variation in the aeroelastic response of continuous tow sheared composite wing. *Composite Structures*, 119706. <https://doi.org/10.1016/j.compstruct.2025.119706>
7. McGurk, M., & Yuan, J. (2025). Prediction and validation of aeroelastic limit cycle oscillations using harmonic balance methods and Koopman operator. *Nonlinear Dynamics*, 113(22), 30841-30868. <https://doi.org/10.1007/s11071-025-11065-8>

8. Zhong, Y., Ma, X., Yuan, J., & Zhang, Z. (2024). Electromechanical modelling and vibration control analysis of cyclic symmetric structures with piezoelectric vibration absorber. *Journal of Vibration & Control*. <https://doi.org/10.1177/10775463241299858>
9. *Wang, X., Yuan, J., Szydlowski, M., & Schwingshackl, C. (2024). Full-field measurement of lightweight nonlinear structures using 3D SLDV. *AIAA Journal*. <https://doi.org/10.2514/1.J064240>
10. *Yuan, J., Gastaldi, C., Denimal Goy, E., & Chouvion, B. (2024). Friction damping for turbomachinery: a comprehensive review of modelling, design strategies, and testing capabilities. *Progress in Aerospace Sciences*, 147, Article 101018. <https://doi.org/10.1016/j.paerosci.2024.101018>
11. *McGurk, M., Lye, A., Renson, L., & Yuan, J. (2024). Data-driven Bayesian inference for stochastic model identification of nonlinear aeroelastic systems. *AIAA Journal*, 62(5), 1889-1905. <https://doi.org/10.2514/1.J063611>
12. *Yuan, J., Szydlowski, M., & Wang, X. (2024). An optimal sparse sensing approach for scanning point selection and response reconstruction in full field structural vibration testing. *Mechanical Systems and Signal Processing*, 212, Article 111298. <https://doi.org/10.1016/j.ymssp.2024.111298>
13. Tufekci, M., Sun, Y., Yuan, J., Maharaj, C., Liu, H., Dear, J. P., & Salles, L. (2024). Analytical vibration modelling and solution of bars with frictional clamps. *Journal of Sound and Vibration*. <https://doi.org/10.1016/j.jsv.2024.118307>
14. *McGurk, M., Yuan, J., & Stodieck, O. (2023). Probabilistic aeroelastic analysis of high-fidelity composite aircraft wings with manufacturing variability. *Composite Structures*. <https://doi.org/10.1016/j.compstruct.2023.117794>
15. Li, Q., Alinsworth, O., Allegri, G., Yuan, J., & Scarpa, F. (2023). Assessing the mechanical and static aeroelastic performance of cellular Kirigami wingbox designs. *Aerospace Science and Technology*, 143, Article 108716. <https://doi.org/10.1016/j.ast.2023.108716>
16. Zhang, W., Neville, R., Zhang, D., Yuan, J., Scarpa, F., & Lakes, R. (2023). Bending of kerf chiral fractal lattice metamaterials. *Composite Structures*, 318, Article 117068. <https://doi.org/10.1016/j.compstruct.2023.117068>
17. Peddakotla, S. A., Yuan, J., Minisci, E., Vasile, M., & Fossati, M. (2023). A numerical approach to evaluate temperature-dependent peridynamics damage model for destructive atmospheric entry of spacecraft. *The Aeronautical Journal*, 127(1309), 398-427. <https://doi.org/10.1017/aer.2022.69>
18. *Yuan, J., Salles, L., Nowell, D., & Schwingshackl, C. (2023). Influence of mesoscale friction interface geometry on the nonlinear dynamic response of large assembled structures. *Mechanical Systems and Signal Processing*, 187, Article 109952. <https://doi.org/10.1016/j.ymssp.2022.109952>

19. Ma, X., Zhong, Y., Cao, P., Yuan, J., & Zhang, Z. (2023). Uncertainty quantification and sensitivity analysis for the self-excited vibration of a spline-shafting system. *Journal of Risk and Uncertainty in Engineering Systems, Part B: Mechanical Engineering*. <https://doi.org/10.1115/1.4063069>
20. Yekai, S., Yaguang, W., Xing, W., Yu, F., Yuan, J., & Dayi, Z. (2022). Review of the Damped Nonlinear Normal Mode in the Analysis of Friction Dampers of Blades/Blisks. *Journal of Aerospace Power*, 37(10), 2167-2187. Advance online publication. <https://doi.org/10.13224/j.cnki.jasp.20220264>
21. Wang, X., Szydlowski, M., Yuan, J., & Schwingshackl, C. (2021). A multi-step interpolated-FFT procedure for full-field nonlinear modal testing of turbomachinery components. *Mechanical Systems and Signal Processing*, 169. <https://doi.org/10.1016/j.ymssp.2021.108771>
22. Zhu, Y.-P., Yuan, J., Lang, Z. Q., Schwingshackl, C. W., Salles, L., & Kadiramanathan, V. (2021). The data driven surrogate model based dynamic design of aero-engine fan systems. *Journal of Engineering for Gas Turbines and Power*, 143(10), Article 101006 . <https://doi.org/10.1115/1.4049504>
23. *Yuan, J., Sun, Y., Schwingshackl, C., & Salles, L. (2021). Computation of damped nonlinear normal modes for large scale nonlinear systems in a self-adaptive modal subspace. *Mechanical Systems and Signal Processing*, 162, Article 108082. Advance online publication. <https://doi.org/10.1016/j.ymssp.2021.108082>
24. *Sun, Y., Yuan, J., Vizzaccaro, A., & Salles, L. (2021). Comparison of different methodologies for the computation of damped nonlinear normal modes and resonance prediction of systems with non-conservative nonlinearities. *Nonlinear Dynamics*, 104, 3077–3107. Advance online publication. <https://doi.org/10.1007/s11071-021-06567-0>
25. *Sun, Y., Yuan, J., Denimal, E., & Salles, L. (2021). Nonlinear modal analysis of frictional ring damper for compressor blisk. *Journal of Engineering for Gas Turbines and Power*, 143(3). <https://doi.org/10.1115/1.4049761>
26. *Yuan, J., Fantetti, A., Denimal, E., Bhatnagar, S., Pesaresi, L., Schwingshackl, C., & Salles, L. (2021). Propagation of friction parameter uncertainties in the nonlinear dynamic response of turbine blades with underplatform dampers. *Mechanical Systems and Signal Processing*, 156, Article 107673. <https://doi.org/10.1016/j.ymssp.2021.107673>
27. *Yuan, J., Schwingshackl, C., Wong, C., & Salles, L. (2021). On an improved adaptive reduced-order model for the computation of steady-state vibrations in large-scale non-conservative systems with friction joints. *Nonlinear Dynamics*, 103, 3283–3300. Advance online publication. <https://doi.org/10.1007/s11071-020-05890-2>
28. *Sun, Y., Vizzaccaro, A., Yuan, J., & Salles, L. (2021). An extended energy balance method for resonance prediction in forced response of systems with non-conservative nonlinearities using damped nonlinear normal mode. *Nonlinear Dynamics*, 103, 3315–3333. <https://doi.org/10.1007/s11071-020-05793-2>

29. Sun, Y., Yuan, J., Pesaresi, L., Denimal, E., & Salles, L. (2020). Parametric study and uncertainty quantification of the nonlinear modal properties of frictional dampers. *Journal of Vibration and Acoustics, Transactions of the ASME*, 142(5). <https://doi.org/10.1115/1.4046953>
30. *Yuan, J., Salles, L., El Haddad, F., & Wong, C. (2020). An adaptive component mode synthesis method for dynamic analysis of jointed structure with contact friction interfaces. *Computers and Structures*, 229, Article 106177. Advance online publication. <https://doi.org/10.1016/j.compstruc.2019.106177>
31. Xie, W., Yang, Y., Meng, S., Peng, T., Yuan, J., Scarpa, F., Xu, C., & Jin, H. (2019). Probabilistic reliability analysis of carbon/carbon composite nozzle cones with uncertain parameters. *Journal of Spacecraft and Rockets*, 56(6), 1765-1774. <https://doi.org/10.2514/1.A34392>
32. *Yuan, J., El-Haddad, F., Salles, L., & Wong, C. (2019). Numerical assessment of reduced order modeling techniques for dynamic analysis of jointed structures with contact nonlinearities. *Journal of Engineering for Gas Turbines and Power*, 141(3), Article 031027 . Advance online publication. <https://doi.org/10.1115/1.4041147>
33. *Yuan, J., Scarpa, F., Titurus, B., Allegri, G., Patsias, S., & Rajasekaran, R. (2017). Novel frame model for mistuning analysis of bladed disk systems. *Journal of Vibration and Acoustics, Transactions of the ASME*, 139(3), Article VIB-15-1430. <https://doi.org/10.1115/1.4036110>
34. *Yuan, J., Scarpa, F., Allegri, G., Titurus, B., Patsias, S., & Rajasekaran, R. (2017). Efficient computational techniques for mistuning analysis of bladed discs: A review. *Mechanical Systems and Signal Processing*, 87(Part A), 71-90. <https://doi.org/10.1016/j.ymssp.2016.09.041>
35. *Yuan, J., Allegri, G., Scarpa, F., Patsias, S., & Rajasekaran, R. (2016). A novel hybrid Neumann expansion method for stochastic analysis of mistuned bladed discs. *Mechanical Systems and Signal Processing*, 72-73, 241-253. <https://doi.org/10.1016/j.ymssp.2015.11.011>
36. Ma, X., Scarpa, F., Peng, H.-X., Allegri, G., Yuan, J., & Ciobanu, R. (2015). Design of a hybrid carbon fibre/carbon nanotube composite for enhanced lightning strike resistance. *Aerospace Science and Technology*, 47, 367-377. <https://doi.org/10.1016/j.ast.2015.10.002>
37. *Yuan, J., Allegri, G., Scarpa, F., Rajasekaran, R., & Patsias, S. (2015). Novel parametric reduced order model for aeroengine blade dynamics. *Mechanical Systems and Signal Processing*, 62-63, 235-253. <https://doi.org/10.1016/j.ymssp.2015.02.015>
38. *Yuan, J., Allegri, G., Scarpa, F., Rajasekaran, R., & Patsias, S. (2015). Probabilistic dynamics of mistuned bladed disc systems using subset simulation. *Journal of Sound and Vibration*, 350, 185. <https://doi.org/10.1016/j.jsv.2015.04.015>
39. *Yuan, J., Allegri, G., & Scarpa, F. (2013). Buffeting mitigation using carbon nanotube composites: A feasibility study. *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, 227(9), 1425-1440. <https://doi.org/10.1177/0954410012461986>